A (Square $n \times n$)

$\underbrace{A\underline{x} = \lambda \underline{x}}$ (non-zero $\underline{x}$, numbers λ).

$A x - \lambda I_n \underline{x} = \underline{0}$

$(A - \lambda I_n)\underline{x} = \underline{0}$ ✓

A has free cols iff $\det(A) = 0$.

If A has free cols; say $\underline{c}_j = \sum_{i<j} \alpha_i \underline{c}_i$

$$\det\begin{bmatrix} | & & | & | & & | \\ \underline{c}_1 & \cdots & \underline{c}_{j-1} & \underline{c}_j & \cdots & \underline{c}_n \\ | & & | & | & & | \end{bmatrix} = \det\begin{bmatrix} | & & | & | & | & & | \\ \underline{c}_1 & \cdots & \underline{c}_{j-1} & \alpha_1\underline{c}_1 + \alpha_2\underline{c}_2 + \cdots + \alpha_{j-1}\underline{c}_{j-1} & \underline{c}_{j+1} & \cdots & \underline{c}_n \\ | & & | & | & | & & | \end{bmatrix}$$

$$= \sum_{i<j} \alpha_i \det\begin{bmatrix} | & & | & | & | & & | \\ \underline{c}_1 & \cdots & \underline{c}_{j-1} & \underline{c}_i & \underline{c}_{j+1} & \cdots & \underline{c}_n \\ | & & | & | & | & & | \end{bmatrix} = 0.$$

If $\det(A) = 0$ then it has free cols.

OR If A has no free cols, then $\det(A) \neq 0$.

$\det(A) = \pm$ product of pivots. $\neq 0$.

depending on how many row exchanges in elimination.

Check $\det(A - \lambda I) = 0$.

$A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$. $A - \lambda I = \begin{pmatrix} 1-\lambda & 0 \\ -1 & 1-\lambda \end{pmatrix}$.

$\lambda I = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$.

$\det(A - \lambda I) = ?$ Use linearity properties.

First row $\underline{(1-\lambda)(1 \quad 0)} + 0(0 \quad 1)$

$$\det(A-\lambda I) = \det\begin{pmatrix} 1-\lambda\,[1 & 0] \\ -1 & (1-\lambda) \end{pmatrix} + \det\begin{pmatrix} 0\,[0 & 1] \\ -1 & 1-\lambda \end{pmatrix}.$$

$$= \underbrace{(1-\lambda)\det\begin{pmatrix} 1 & 0 \\ -1 & 1-\lambda \end{pmatrix}} + \cancel{0\det\begin{pmatrix} 0 & 1 \\ -1 & 1-\lambda \end{pmatrix}}.$$

Second row: $-1\begin{bmatrix} 1 & 0 \end{bmatrix} + (1-\lambda)\begin{bmatrix} 0, & 1 \end{bmatrix}$

$$(1-\lambda)\det\begin{bmatrix} 1 & 0 \\ -1[1\ 0] + (1-\lambda)[0\ 1] \end{bmatrix} = (1-\lambda)\det\begin{bmatrix} 1 & 0 \\ -1[1 & 0] \end{bmatrix} + (1-\lambda)\det\begin{bmatrix} 1 & 0 \\ (1-\lambda)[0. & 1] \end{bmatrix}$$

$$= (1-\lambda)(-1)\underbrace{\cancel{\det\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}}}_{0} + (1-\lambda)^2\underbrace{\det\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{1}.$$

$$= \underbrace{(1-\lambda)^2}.$$

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}. \qquad \det(A-\lambda I) = \underbrace{(1-\lambda)}^2.$$

roots of $\det(A-\lambda I) = 0$ are $1, 1$.

$$A - 1I = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} - 1\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix}.$$

$$\text{null}(A-I) = \left\{ \alpha\begin{pmatrix} 0 \\ 1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}.$$

$$\text{rref}(A-I) = \text{rref}\begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

$\underline{c2} = 0\,\underline{c1}$

$\underline{c2} - \underline{0}\,\underline{c1} = \underline{0}$

characteristic polynomial.

Roots of $\underline{\det(A-\lambda I) = 0}$: Eigenvalues.

If a root is repeated k times in $\det(A-\lambda I) = 0$, we say that eigenvalue has <u>algebraic</u> multiplicity k.

Square Symmetric real A:

$A^T = A$ eg: $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$.

$$A - \lambda I = \begin{pmatrix} 1-\lambda & -1 \\ -1 & 1-\lambda \end{pmatrix}.$$

$\det(A - \lambda I) = (1-\lambda)^2 - 1 = \lambda^2 - 2\lambda + 1 - 1 = \lambda^2 - 2\lambda.$

Roots: $0, 2$.

$\text{null}(A - 0I) = \text{null}(A) = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}.$

$\text{rref}(A) = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}.$ $\underline{c}2 = -\underline{c}1$, $\underline{c}2 + \underline{c}1 = \underline{0}$.

$\text{null}(A - 2I) = \text{null}\left[\begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \right] = \left\{ \alpha \begin{pmatrix} 1 \\ -1 \end{pmatrix} : \alpha \in \mathbb{R} \right\}.$

Dot product of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0$.

The eigenvectors (one from each of null (A) and null $(A - 2I)$) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ form a max linearly indpt set in $\mathbb{R}^2$.

For all $\underline{x} \in \mathbb{R}^2$, $\underline{x} = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$$\begin{aligned} A\underline{x} &= A\left[\alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right] \\ &= A\left[\alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right] + A\left[\beta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right] \\ &= \alpha A \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \beta A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \\ &= \alpha . 0 . \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 2\beta \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \end{aligned}$$